

Setup Guide

Local Attribution Sandbox Setup

This setup guide will help you install the required tools, configure the project locally, and run the attribution workflow step-by-step.

The sandbox is intentionally lightweight and designed to run fully on your local machine using:

- Python
- DuckDB
- Pandas
- CSV datasets

No cloud infrastructure or paid tooling is required.

1. Install Python

Download and install Python:

None

<https://www.python.org/downloads/>

Beginner Setup Notes

If you have never used Python before, don't worry — this project is designed to run locally with only a few setup steps.

Python is the programming language used to run all scripts inside this sandbox.

When Python is installed correctly, your computer can execute `.py` files directly from the terminal or command prompt.

During installation, you may see an option that says:

None

Add Python to PATH

Make sure this option is checked.

This allows your computer to recognise Python commands globally from the terminal.

If Python is not added to PATH correctly, you may see errors like:

None

python is not recognized

or:

None

command not found

These errors simply mean your system cannot locate the Python installation yet.

After installation, restart your terminal or command prompt before continuing.

You can then verify Python installed correctly by running:

None

python --version

If successful, you should see something similar to:

None

Python 3.x.x

2. Clone or Download the Repository

If using GitHub:

None

```
git clone https://github.com/ScaleK2/local-attribution-sandbox.git
cd local-attribution-sandbox
```

Or manually download and extract the ZIP file.

3. Create a Virtual Environment (Optional / Recommended)

A virtual environment keeps project dependencies isolated from the rest of your system.

Windows

None

```
python -m venv venv
venv\Scripts\activate
```

Mac / Linux

None

```
python3 -m venv venv
source venv/bin/activate
```

If successful, your terminal should now show:

None

```
(venv)
```

4. Install Required Packages

Inside the project folder, run:

```
None  
pip install duckdb pandas
```

Or install using:

```
None  
pip install -r requirements.txt
```

5. Verify Project Structure

Your project should look similar to this:

```
None  
local-attribution-sandbox/  
|  
├─ data/  
|   ├── ga4_events.csv  
|   ├── google_ads_clicks.csv  
|   ├── meta_ads_insights.csv  
|   ├── backend_outcomes.csv  
|   ├── data_dictionary.md  
|   ├── simulation_summary.csv  
|   └── master_user_simulation.csv  
|  
├─ scripts/  
|   ├── 01_stitch_data.py  
|   ├── 02_validate_data.py  
|   ├── 03_check_backend_truth_vs_stitched.py  
|   ├── 04_backend_first_conversion_journeys.py  
|   └── 05_build_touchpoint_attribution.py
```

```
|   |— 06_linear_attribution.py
|   |— 07_position_based_attribution.py
|   |— 08_time_decay_attribution.py
|   |— 09_compare_attribution_models.py
|
|— outputs/
|
|— docs/
|   |— setup_guide.pdf
|   |— attribution modelling walkthrough.pdf
|
|— requirements.txt
|— README.md
```

6. Check Dataset Files

Confirm the following CSV files exist inside the `data/` folder:

None

```
data/ga4_events.csv
data/backend_outcomes.csv
data/google_ads_clicks.csv
data/meta_ads_insights.csv
data/master_user_simulation.csv
data/simulation_summary.csv
```

These datasets simulate:

- frontend analytics events,
 - advertising platform interactions,
 - and backend conversion outcomes.
-

7. Run The Workflow Sequentially

Each script builds on the previous step.

Run them in order.

Example:

None

```
python scripts/01_stitch_data.py
```

Continue sequentially:

None

```
python scripts/02_validate_data.py
python scripts/03_check_backend_truth_vs_stitched.py
python scripts/04_backend_first_conversion_journeys.py
python scripts/05_build_touchpoint_attribution.py
python scripts/06_linear_attribution.py
python scripts/07_position_based_attribution.py
python scripts/08_time_decay_attribution.py
python scripts/09_compare_attribution_models.py
```

8. Follow Along With The Walkthrough Guide

Open:

None

```
docs/attribution modelling walkthrough.pdf
```

The walkthrough guide contains:

- screenshots,
- explanations,
- commentary,
- behavioural interpretation,
- and strategic observations for each step.

The recommended workflow is:

1. Run the script
2. Observe the output
3. Compare against the walkthrough
4. Reflect on the behavioural implications

Troubleshooting

Python Not Found

If you see:

None

```
python is not recognized
```

Make sure Python was added to your system PATH during installation.

Missing Package Errors

Install dependencies again:

None

```
pip install duckdb pandas
```

CSV File Errors

Confirm:

- all datasets exist,
 - filenames match exactly,
 - and files are inside the `data/` folder.
-

Final Note

This sandbox is intentionally educational rather than production-grade.

The goal is not building a perfect attribution system.

The goal is understanding:

- how attribution systems work,
- how customer journeys are reconstructed,
- and how different attribution assumptions shape marketing interpretation.